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5 April 1973
SUPERSEDING
MIL-C-915/7C
1 August 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE DSWS

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction (Watertight)

First - Copper conductor, Navy Standard size 4(7), coated
Second - Synthetic rubber insulation. Nominal wall thickness 0.073 inch. Standard Identification Code, applied by Method 3. An additional covering of clear polyamide may be employed at the manufacturer's option.
Third - Two conductors shall be cabled together with a lay not greater than 5 inches. Fillers of synthetic rubber like compound shall be used to attain a firm, well rounded cross section. Fillers may be laid in the valleys during the cabling operation or may be extruded into the valleys separately. Fillers shall bbe readily separable from the insulation without damage thereto and without destroying the conductor identification.
Fourth - Binder tape over the conductor assembly, at manufacturer's option.
Fifth - Shielding braid of AWG No. 34, tin-coated copper, with an angle of 30 to 40 degrees and minimum coverage of 88 percent.
Sixth - An overall jacket (black), nominal wall thickness 0.060 inch. If needed to meet electrical tests, the jacket may comprise an inner layer of rubber insulation, such as synthetic rubber and an outer layer of polychloroprene. The jacket shall be readily separable from the underlying shield, leaving the shield adequately clean for soldering. Cable surface marking required.

Details

Type and style	Overall diameter	
	Minimum (inch)	Maximum (inch)
DSWS-4	0.770	0.800

EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
Conductor resistance - Ohms/1000 ft. at 25°C., maximum	2.57
Voltage withstand - Volts, rms, minimum	
Conductor to conductor (5 minutes)	10,000
Conductor to shield (5 minutes)	10,000
Shield to water (5 minutes)	500
Voltage withstand - Volts, d.c., minimum	
Conductor to conductor (15 minutes)	25,000
Conductor to shield (15 minutes)	25,000
Shield to water (15 minutes)	1,500

(D) Denotes changes.

FSC 6145

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	Requirements:
Basic electrical (Cont'd):	
Insulation resistance - Megohms/1000 ft., minimum	
Shield to water	100
Conductor to conductor	200
Conductor to shield	200
Group A:	
Visual and dimensional	
Hydrostatic (open end) - 1000 psi, cu. in., 2 hrs, maximum	0
Mutual capacitance - pF/ft. at 1 KHz	40±10
Group B:	
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - psi, minimum	600
Elongation - Percent, minimum	350
Jacket (cable)	
Tensile strength - psi, minimum	1,800
Elongation - Percent, minimum	300
Set - Inch, maximum	3/8
Group C:	
Physical (aged)	
Insulation (conductor)	
Air pressure heat	
Tensile strength - Percent of unaged, minimum	70
Elongation - Percent of unaged, minimum	75
Jacket (cable)	
Air pressure heat	
Tensile strength - Percent of unaged, minimum	50
Elongation - Percent of unaged, minimum	50
Hot oil immersion	
Tensile strength - Percent of unaged, minimum	60
Elongation - Percent of unaged, minimum	60
Permanence of printing (jacket) cycles, minimum	250

(D)

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical, all of Group A, B and C plus the following:

Insulation resistance per 1000 feet at room temperature from shield to water, 50 megohms, minimum after 90 days continuous submergence in water; values recorded at 1-week intervals.

Flexing endurance - 1000 cycles of bending a specimen, 4-inch maximum diameter mandrel at minus 40° + 2°C. Upon successful completion of this test, specimen shall meet the voltage withstand test and the hydrostatic (open end) test.

UNIT ORDERING LENGTHS:
500 feet (Nominal) Coils

Custodians:
Army - MI
Navy - SH
Air Force - 80
Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80
User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
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